

RUNOFF storm episodes

A dataset for flash flood detection models

What the dataset contains, how it was built, how to use the notebook, and what the radar coverage layer means



Symbols used on the website: squares are episodes, circles are reports, triangles are gages

Two ways to access RUNOFF

Both are on the same website. These slides describe the second one.

NextGen path (built at the Summer Institute)

- Starts from the stream gage: USGS basins under 1000 km²
- Flash flood events found in the gage record and matched to Storm Events
- MRMS radar rainfall or AORC forcing at the chosen time step
- Packages ready to run in NextGen
- A gage in the basin is required

Storm episodes path (built after the Institute)

- Starts from the storm reports. No gage is required
- Every weather-caused flash flood episode in the lower 48 states, water years 2021 to 2025
- Reports, watersheds, rainfall, return periods, FLASH model output, guidance, gages, radar coverage
- One folder of flat tables and a notebook
- Built for flash flood guidance style detection models

Website: nwc-cuahsi-summer-institute.github.io/runoff

Dataset folder

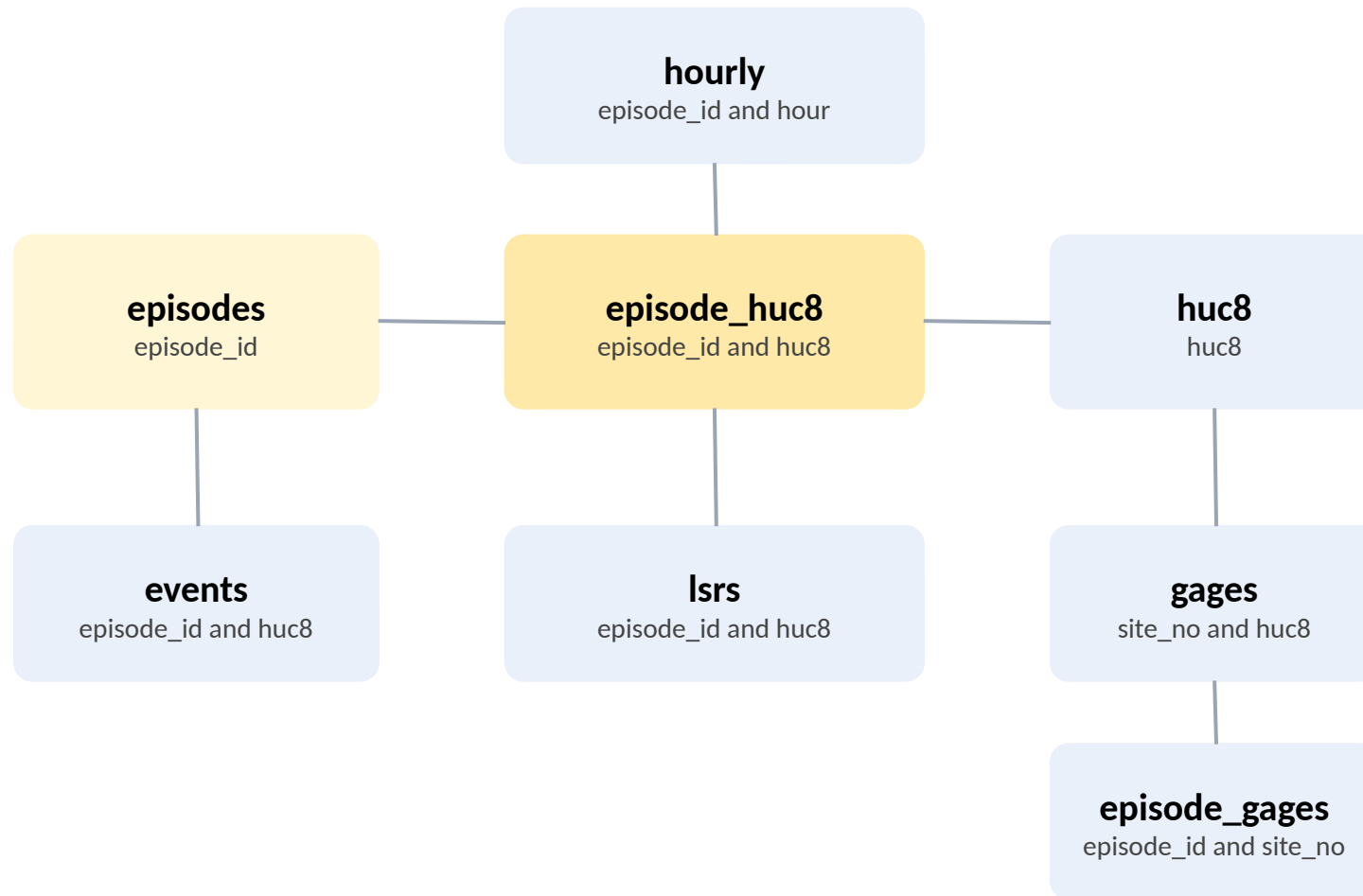
The folder storm_episodes in the RUNOFF repository on GitHub

```
storm_episodes/  
  README.md  
  DATA_DICTIONARY.csv  
  RUNOFF_storm_episodes_overview.pdf  
  data/  
    episodes.parquet      episodes.csv  
    episode_huc8.parquet  episode_huc8.csv  
    events.parquet        events.csv  
    lsrs.parquet          lsrs.csv  
    hourly.parquet        hourly.csv  
    huc8.parquet          huc8.csv  
    gages.parquet         gages.csv  
    episode_gages.parquet episode_gages.csv  
    radar_cov.tif  
  notebooks/  
    quickstart.ipynb
```

- README.md: what the tables are and how the numbers were computed
- DATA_DICTIONARY.csv: every column with its type and meaning
- RUNOFF_storm_episodes_overview.pdf: these slides
- data/: eight tables, each as Parquet and CSV. Parquet keeps the column types. CSV opens in Excel
- radar_cov.tif: the radar coverage layer, 4 MB
- notebooks/quickstart.ipynb: a guided tour of the tables, already executed, figures included
- Total size about 55 MB. Clone the repository, download the folder, or download the zip from the latest release
- The tables can be read with pandas

The eight tables and how they join

episode_id links every table to the episodes. huc8 links the watershed tables.



What one row is

episodes: one storm episode, with its time window, impacts, footprint, rainfall, return periods, FLASH output, guidance and narrative

episode_huc8: one watershed inside one episode. This is the modeling table. It carries the label flooded

events, lsrs: one report, with its narrative and the watershed it fell in

hourly: one hour of one episode: rain, return period and FLASH products over the footprint

huc8: one watershed: name, area, gages, radar coverage, number of episodes with reports

gages, episode_gages: USGS gages under 1000 km2 and the episode footprints that contain them

Dataset summary

Lower 48 states and DC, water years 2021 to 2025, weather-caused flash floods only

5,418

storm episodes

20,882

Storm Events flash flood reports

24,907

Local Storm Reports matched

35,697

watershed-episodes, 32 % reported flooded

1,961

HUC8 watersheds

5,897

USGS gages under 1000 km2

These episodes include 564 fatalities and 9.5 billion dollars of reported damage. Every episode has rainfall, return period, FLASH and guidance statistics. 5,239 episodes have at least one gage in the footprint.

How the dataset was built

To build this dataset the following 9 steps were followed. All results are precomputed and stored as files.

1 Episodes and events

Flash flood reports from the NOAA Storm Events Database for the lower 48 states. Dam and levee failures removed. Times converted to UTC.

2 Local Storm Reports

NWS Local Storm Reports from the Iowa Environmental Mesonet archive, matched to episodes by time window and county.

3 Watershed footprint

Every HUC8 watershed (USGS WBD) touched by the reporting counties or containing a report.

4 Rainfall

MRMS QPE Pass 2, hourly, 1 km. Rolling 1, 3 and 6 hour maxima and episode totals for every watershed.

5 Return periods

FLASH QPE ARI for 1, 3 and 6 hours: the return period of the rainfall at each cell, and the share of the area above 1 to 200 years.

6 FLASH models and guidance

CREST, SAC-SMA and hydrophobic peak unit streamflow. Ratio of MRMS rainfall to flash flood guidance.

7 Gages

USGS gages under 1000 km² with NWIS names and NLDI drainage basins, placed in the footprints.

8 Radar coverage

The hydrologic radar coverage layer summarized per watershed and sampled at every gage and report.

9 Export

All results joined into eight flat tables, a data dictionary and a README.

Data sources

Content	Source	Used for
Flash flood episodes, events, impacts, narratives	NOAA NCEI Storm Events Database	episodes, events, the label, fatalities and damage
Local Storm Reports with remarks	NWS Local Storm Reports, Iowa Environmental Mesonet archive	lsrs, the label, report counts per watershed
Hourly rainfall, 1 km	MRMS MultiSensor QPE Pass 2 (Iowa State mirror, AWS as fallback)	rain totals, rolling maxima, hourly series
Return periods	NSSL FLASH QPE ARI 1, 3 and 6 hour	peak return period and area above 1 to 200 years
Model output and guidance	NSSL FLASH: CREST, SAC-SMA, hydrophobic; QPE to FFG ratio	peak unit streamflow, share of the area above guidance
Watersheds	USGS Watershed Boundary Dataset, HUC8	footprints, per watershed statistics, maps
Gages and basins	USGS GAGES-II, NWIS, NLDI	gages table, basin polygons, coverage at the gage
Radar coverage	radar_cov.tif, hydrologic coverage on the 0.01 degree grid	coverage per watershed, at gages and at reports

All sources are public. The processing scripts are in the engine folder of the repository.

The modeling table: episode_huc8

One row is one watershed inside one storm episode. 35,697 rows, 74 columns.

- **Features, by group**
- rain: episode total and rolling 1, 3 and 6 hour maxima (mm)
- ari: peak return period (years) and share of the area above 1 to 200 years
- crest, sac, hp: FLASH peak unit streamflow (m3/s/km2)
- ffg: ratio of rainfall to flash flood guidance (1.0 means the rain reached the guidance)
- n_gages, huc8_area_km2, radar_coverage columns
- **Label**
- flooded = 1 when any Storm Events report or LSR fell inside the watershed during the episode
- 32 % of the rows are positive
- **Notes on the label**
- Reports depend on where people and roads are
- A value of 0 means no report was filed, not that no flooding occurred

Four rows of one episode (southern Missouri, April 2025)

huc8	watershed	flood	events	LSRs	1 h peak mm	6 h area >= 10 yr	CREST	radar
11010010	Spring	1	12	12	56	17 %	8.8	0
11010011	Eleven Point	1	22	21	55	3 %	5.8	4
11010006	North Fork White	1	13	14	45	2 %	8.3	47
11010007	Upper Black	0	0	0	62	2 %	7.5	18

Upper Black had the highest 1 hour rainfall of the four and no report. The radar coverage is 18 % and the watershed has few gages. This row shows why the radar and gage columns are in the table: a watershed with no report may be poorly observed rather than dry.

The quickstart notebook

notebooks/quickstart.ipynb: a guided tour in 14 short sections

1 Load the tables

2 Read the data dictionary

3 One episode from start to end

4 Filtering examples

5 Summaries by year, state and month

6 Maps without a GIS library

7 The narratives

8 Timing inside an episode

9 The modeling table and scores

10 The radar layer for one basin

11 Labels and negatives

12 Gages: link to the NextGen path

13 GIS export

14 Saving subsets, notes

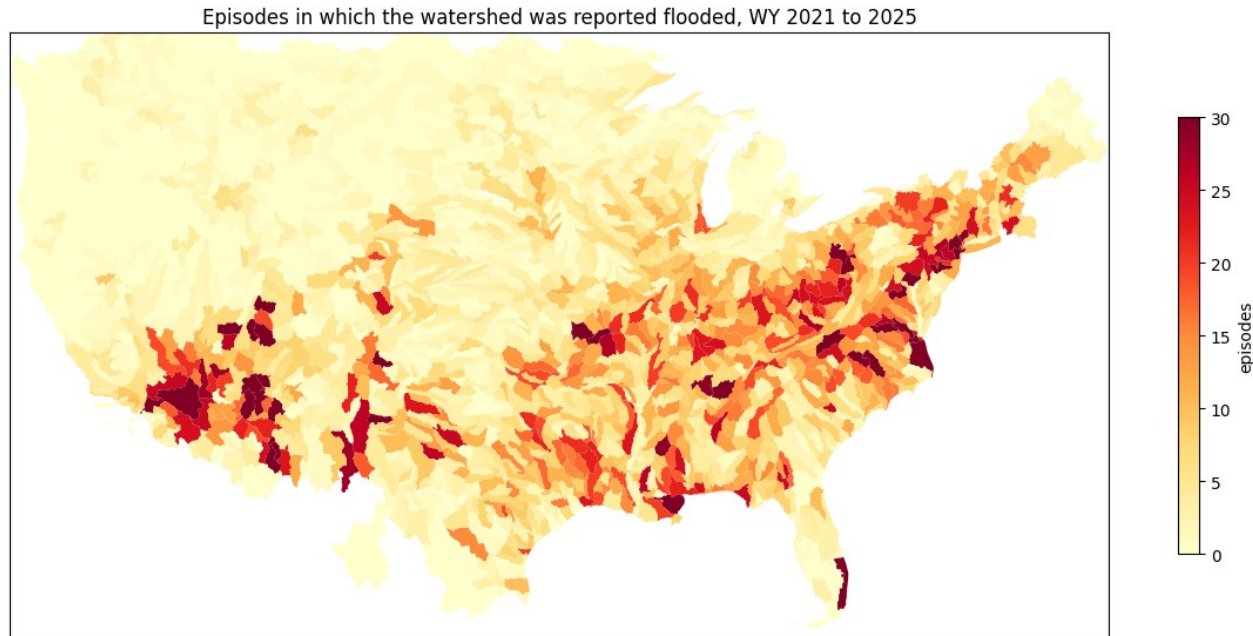
What to expect

- Runs from top to bottom in about two minutes
- Every code cell has a short note before it that explains what it does and why
- Needs pandas and matplotlib. scikit-learn and rasterio are optional; the cells that use them are skipped if they are missing
- Outputs are saved in the file, so GitHub shows the tables and figures without running anything
- Any cell can be copied into your own script. The tables are plain pandas DataFrames

Example figures from the notebook

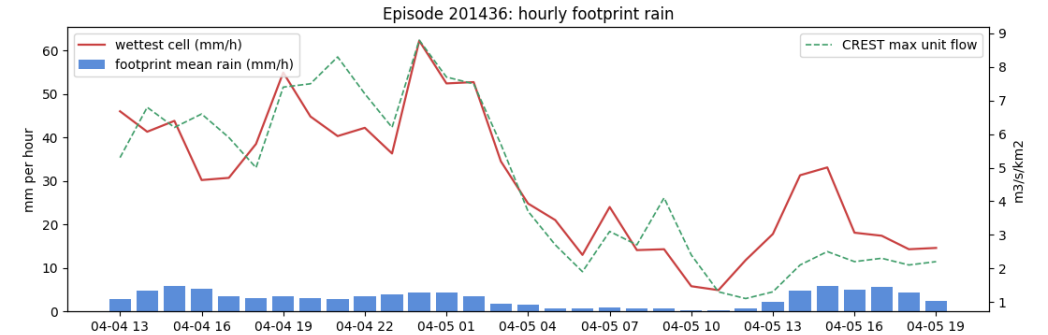
Three of the eight figures

Watersheds reported flooded, water years 2021 to 2025

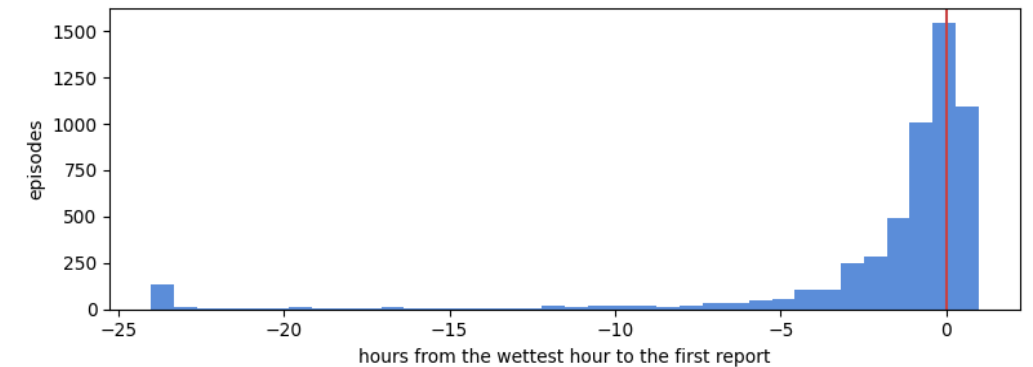


The median first report comes half an hour before the wettest hour. 86 % of first reports fall between 3 hours before and 6 hours after it.

One episode hour by hour: mean rain over the footprint, wettest cell, CREST peak



Time of the first report relative to the wettest hour



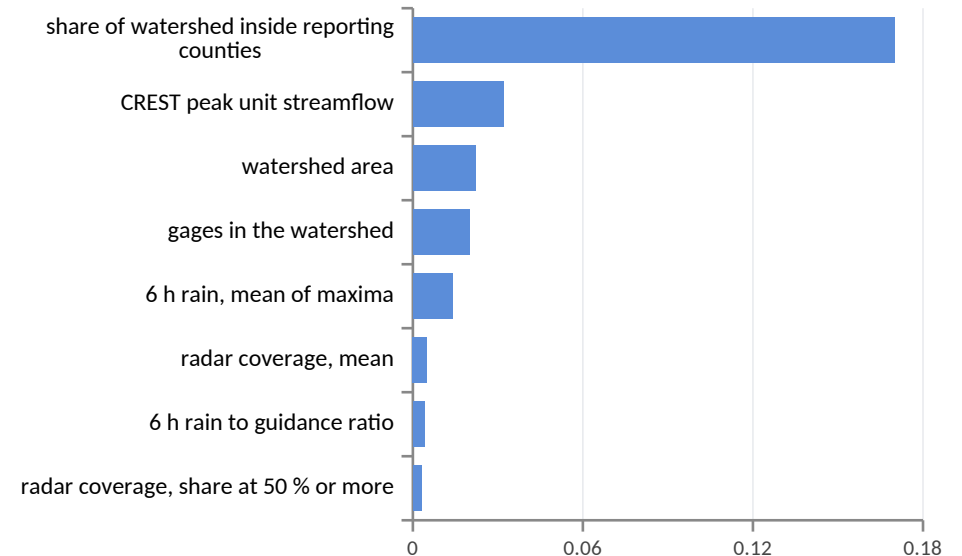
Baseline rules and a first model

Trained on water years 2021 to 2024, tested on water year 2025. One row is one watershed in one episode.

Method	POD	FAR	CSI
Rain reached the guidance somewhere in the watershed	0.87	0.54	0.43
Any cell reached a 2 year, 6 hour rainfall	0.86	0.55	0.42
CREST peak unit streamflow of 2 or more	0.88	0.50	0.47
Gradient boosting on all columns, threshold 0.3	0.85	0.34	0.59

- POD: share of the flooded watersheds that were detected
- FAR: share of the detections that were false
- CSI: hits divided by hits plus misses plus false alarms
- The model keeps the detection rate of the best rule and lowers the false alarm ratio from 0.50 to 0.34
- These scores are the baseline for models trained on this dataset

Most important columns in the model

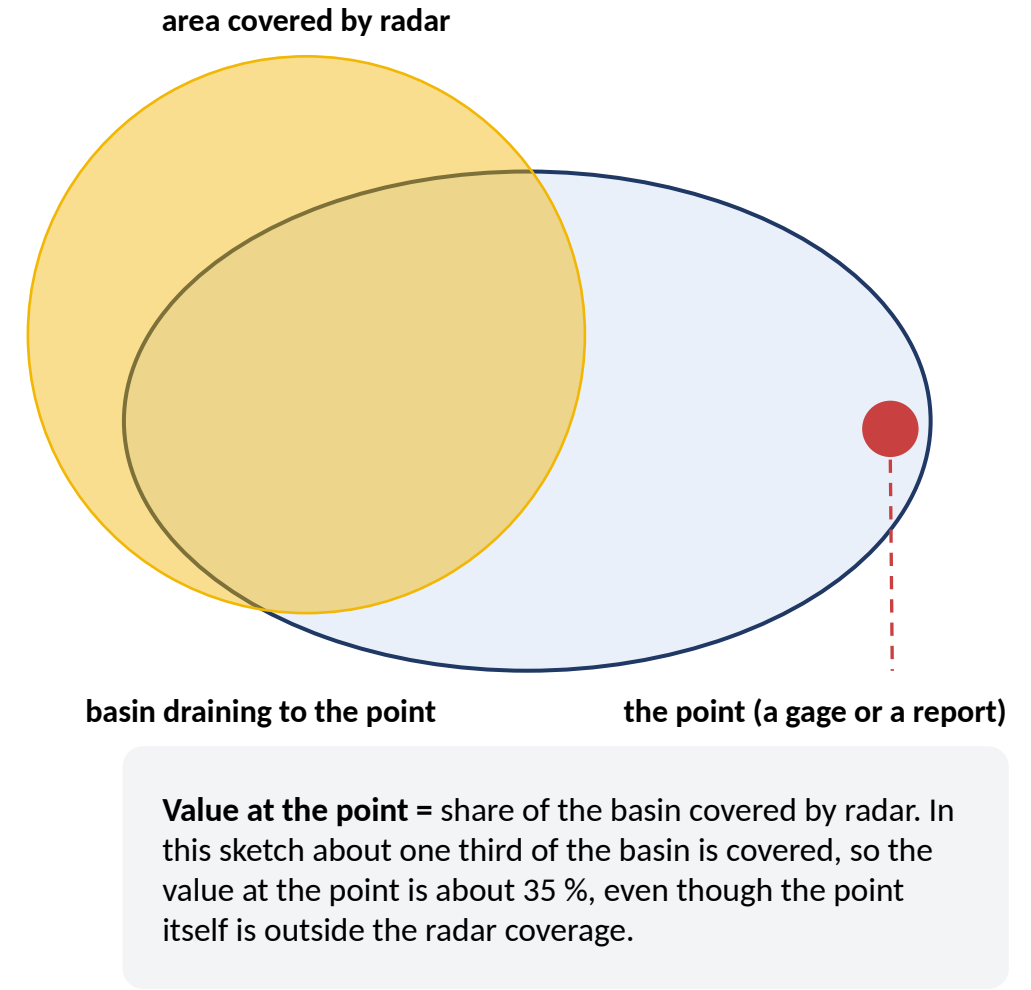


Drop in ROC AUC on the test year when one column is shuffled. The share inside reporting counties comes from the reports themselves and should be used with care. CREST, watershed area, gage count and 6 hour rain are the next most important.

The radar coverage layer

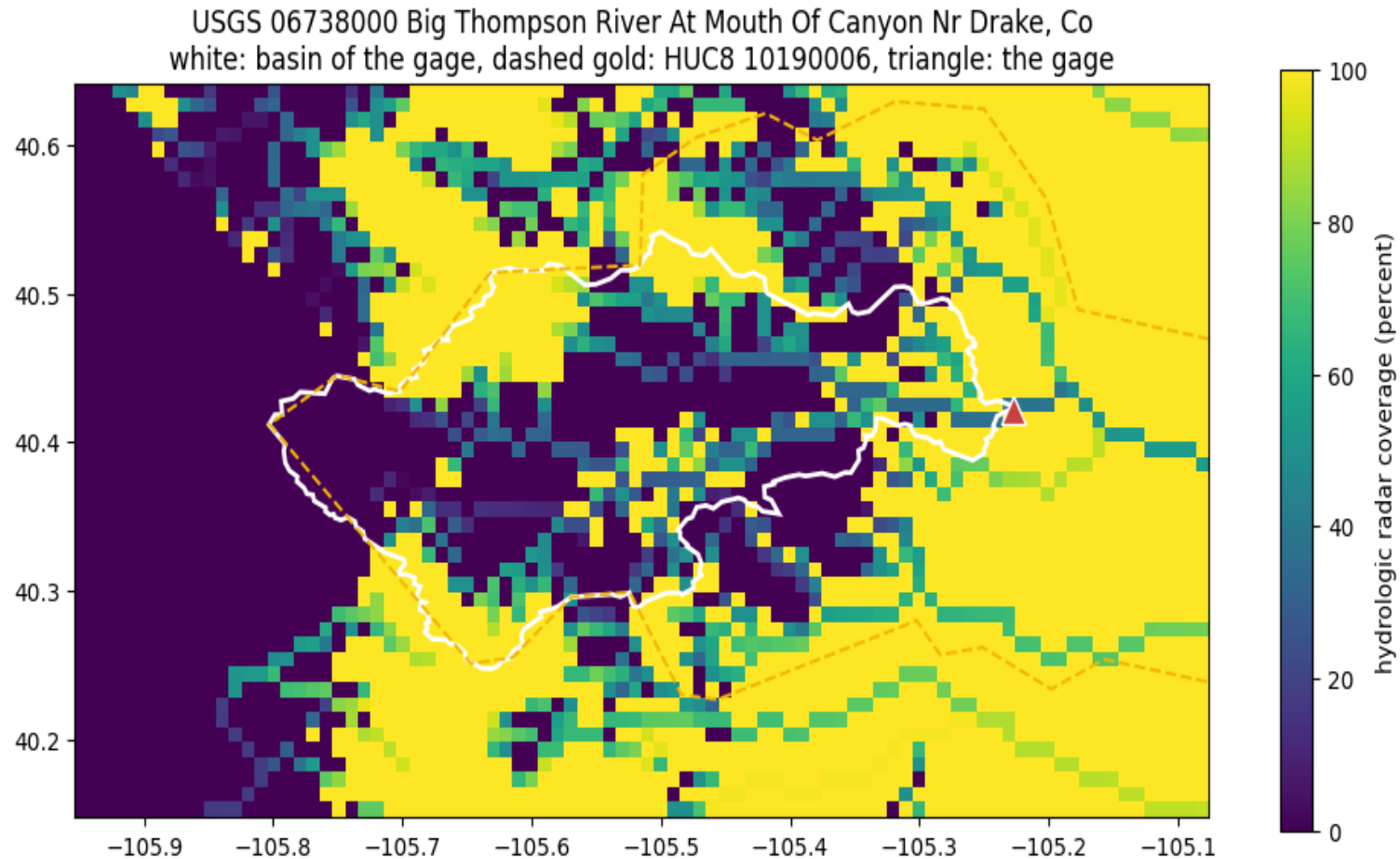
radar_cov.tif is a hydrologic coverage, not a plain radar coverage map

- A grid at 0.01 degree, the same grid as MRMS, with values from 0 to 100
- **At each pixel, the value is the percent of the drainage area upstream of that pixel that is covered by weather radar**
- The value at a gage is the radar coverage of the whole basin above the gage
- The value at a flash flood report is the radar coverage of the basin draining to that location
- A headwater pixel represents a small area. A pixel at a basin outlet represents the whole basin, so low coverage in the headwaters lowers the value downstream
- **The layer is used in three ways in the dataset**
 - at each gage: radar_coverage_at_gage
 - at each report: radar_coverage_at_report, and its mean per watershed and episode
 - per watershed: mean, minimum, maximum, share of the area at 50 % or more and at 100 %



Example: Big Thompson River basin

USGS 06738000, Big Thompson River at the mouth of the canyon near Drake, Colorado. 791 km², 12 RUNOFF events.



- White line: the basin draining to the gage. Dashed gold line: its HUC8. Triangle: the gage
- Dark pixels: little or no radar coverage upstream. Yellow pixels: full coverage
- **Value at the gage: 38 %**
- Mean of the pixels inside the basin: 39 %
- Mean over the whole HUC8: 66 %, with cells from 0 to 100 %
- The upper basin has no radar coverage. The value at the gage reflects this. The HUC8 mean does not
- This is why the dataset samples the layer at gages and reports, and not only per watershed

Radar coverage and flood reports

For similar rainfall, how often was flooding reported?

Watershed-episodes with a 1 hour peak rainfall of at least 40 mm, grouped by radar coverage

watershed mean coverage	watershed-episodes	reported flooded	mean 1 h peak rain	gages per watershed
0 to 10 %	1,024	29 %	55 mm	0.8
10 to 50 %	2,279	38 %	57 mm	2.1
50 to 90 %	3,810	49 %	57 mm	3.4
90 to 100 %	8,581	54 %	58 mm	4.6

What the table shows

- For the same rainfall, watersheds with low radar coverage are reported flooded about half as often as watersheds with full coverage

Points to keep in mind

- Fewer people and roads means fewer reports. Gage density follows the same pattern
- Radar underestimates rainfall where it has no coverage, so the measured rain is lower there
- The coverage columns allow a model to account for this difference

How to access the dataset and next steps

Access

- Folder: github.com/NWC-CUAHSI-Summer-Institute/runoff, folder `storm_episodes`
- Zip: the latest release of the repository
- Website: episodes.html to browse, filter and download one episode at a time
- Start with [notebooks/quickstart.ipynb](#)

Suggested use

- Filter episodes by impacts, rainfall, return periods, FLASH output, guidance, gages and radar coverage
- Train on `episode_huc8` and test on the last water year
- Compare results with the baseline CSI of 0.59 at the watershed level

Planned additions

- Watershed-hours without flooding, so the label also covers storms where nothing was reported
- HUC12 watersheds, closer to the scale of flash flood guidance
- MRMS radar quality index and beam height per watershed, next to the coverage layer
- Soil moisture before each episode